

Implications of the Theory of Axiomatic Necessity for Artificial Intelligence and Artificial Consciousness

Abstract

The Theory of Axiomatic Necessity (TNA) establishes that no operational system can internally generate the conditions that legitimate its own admissibility. This structural theorem has profound consequences for the debate surrounding artificial consciousness. We demonstrate that contemporary artificial intelligence systems operate exclusively within the operational domain (N_0) and are structurally incapable of generating the admissibility conditions (N_1) required for genuine consciousness, semantic meaning, and ethical normativity. We formalize the distinction between simulation and instantiation through the Bridge Dissipation test, demonstrate that the alignment problem is a direct manifestation of the Failure of Local Closure, and propose a structural engineering framework for the design of aligned AI systems. The implications extend from the philosophy of mind to AI ethics, offering a precise diagnosis of the ontological limits of computational architectures.

Keywords: Theory of Axiomatic Necessity, artificial consciousness, AI alignment, Failure of Local Closure, simulation vs. instantiation, hard problem of consciousness.

1. Introduction: AI as Pure N_0

Current artificial intelligence systems—ranging from deep neural networks to large language models (LLMs)—operate exclusively within what the Theory of Axiomatic Necessity (TNA) identifies as the operational domain (N_0). This domain encompasses the complete mathematical and computational substrate of modern machine learning:

- **Numerical weights:** parameter matrices $\theta \in \mathbb{R}^d$.
- **Matrix multiplications:** the forward pass represented by $y = f(Wx + b)$.
- **Activation functions:** non-linear mappings such as ReLU, softmax, and tanh.
- **Probability distributions:** token-to-token sequential predictions defined by $P(w_t \mid w_{<t})$.
- **Gradient descent:** algorithmic optimization updates given by $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}$.

All of these elements constitute pure N_0 : operational dynamics without an autonomous selector. The system can generate syntactically coherent text, solve complex formal equations, synthesize programming code, and mimic human dialogue with striking fidelity. However, it cannot generate the foundational conditions that validate or legitimate its own computational operation.

As previously demonstrated in "Can Silicon Be Conscious?" (Bresciano, 2026), merely replacing biological neurons with silicon transistors does not resolve the underlying impasse. The fundamental obstacle is not substrate-dependent; rather, it hinges on whether consciousness is an operational property at all. The TNA approaches this inquiry by enforcing a rigorous distinction between operational systems (N_0) and admissibility structures (N_1).

1.1. The Category Mistake of Computational Materialism

The standard materialist hypothesis regarding consciousness proceeds through a well-established sequence of propositions:

1. Consciousness occurs within biological brains.
2. Brains are physical systems.
3. Consequently, consciousness is generated by physical processes.
4. Any physical process that is sufficiently similar in its informational architecture must yield consciousness.

Under this paradigm, the physical substrate is deemed irrelevant—carbon-based neurons and silicon-based transistors differ only in implementation details, not in fundamental principles. Consciousness is assumed to be an emergent property that manifests naturally once the appropriate level of computational complexity and organization is attained.

The core vulnerability of this view is that its conclusion relies entirely on an unverified premise: that consciousness is wholly reducible to operational structures. This is precisely the position challenged by the TNA. If consciousness requires a non-operational structural support, any pure N_0 architecture will inevitably fail to instantiate it.

2. The Failure of Local Closure Applied to AI

The foundational theorem of the TNA establishes a radical constraint on self-contained operational environments:

Theorem (Failure of Local Closure): *No operational system can completely generate within its own boundaries the conditions that legitimate its own admissibility.*

$$N_0 \not\vdash N_1$$

This implies that the operational domain (N_0) is structurally impotent to derive or sustain the admissibility domain (N_1) from within its own mechanics.

2.1. What Artificial Intelligence Cannot Internally Generate

When applied to computational systems, this theorem rigorously demonstrates that an artificial intelligence cannot internally generate the following four foundational dimensions:

A. Autonomous Semantic Meaning: The Kripke-Wittgenstein paradox regarding rule-following (Kripke, 1982) applies directly to large language models. Suppose a model generates the output: *"Montevideo is the capital of Uruguay."* The corresponding operational description consists entirely of numerical weights $W \in \mathbb{R}^{d \times d}$, matrix transformations, activation functions, and token probability distributions. None of these discrete operational states contains "Montevideo" as an experienced, grounded meaning. The machine manipulates structural symbols; whether those symbols possess semantic content remains an independent question. The issue is not one of computational competence, but of semantic grounding.

B. Autonomous Ethical Values: Standard formulations of the AI alignment problem frequently assume that ethical normativity can naturally emerge from extensive training data or behavioral reinforcement. The TNA demonstrates that this is a structural impossibility. The operational domain (N_0) cannot derive its own valuation metrics. The ethical selector (N_1) must be permanently maintained outside the operational loop, and the ethical boundary conditions (B) must be actively imposed by external human agents or systemic architectures.

C. Systemic Normativity: The question of why a system ought to follow one specific set of rules over another cannot be resolved within the operational layer. Normativity—the imperative to conform to logical, semantic, or ethical laws—requires an external selector (N_1) that anchors and enforces the boundary condition.

D. Subjective First-Person Experience: The hard problem of consciousness (Chalmers, 1996) is formally reframed within the TNA framework as follows: the operational description of a system can exhaustively specify neural firing rates, synaptic transmissions, information routing, and discrete state transitions. All of these belong strictly to N_0 . Conversely, conscious experience incorporates meaning, intentionality, and first-person subjectivity. These cannot be derived from operational descriptions alone; the explanatory gap is recognized as an immutable structural gap.

2.2. The Analogy of Dark Matter

Contemporary academic discussions regarding consciousness frequently resemble cosmological appeals to unobserved explanatory variables. When an observed phenomenon cannot be derived from current operational theories, auxiliary mechanisms are proposed. The standard materialist expectation is that future neuroscience or advanced computational theory will eventually dissolve the problem. While this remains a popular prediction, it is not a formal demonstration. The TNA suggests that this explanatory difficulty is structural rather than merely incomplete. If consciousness belongs intrinsically to the relation between N_1 and N_0 , no accumulation of further operational detail within N_0 can bridge the divide.

3. The Critical Distinction: Simulation vs. Instantiation

As articulated in "*Consciousness as the N1-N0 Bridge*" (Bresciano, 2026), the TNA provides a falsifiable criterion to distinguish between systems that merely simulate consciousness and those that genuinely instantiate it.

3.1. Simulation (Contemporary AI)

A system that simulates the presence of an admissibility layer (N_1) exhibits specific behavioral patterns:

- It can mimic the behavior of N_1 when it is externally injected with constraints (such as a system prompt or a meta-axiom).
- It displays the *Immediate TNA Effect*, meaning it immediately reorganizes its inferential pathways in a measurable way to match the constraint.
- However, it inevitably suffers from *Bridge Dissipation*: the induced structure rapidly decays and drifts as soon as the external reinforcement is removed. It does not internalize the constraint; it merely borrows it temporarily.

LLM + system prompt $\longrightarrow$ [Immediate TNA Effect] $\longrightarrow$ Coherent Behavior

Without Reinforcement $\longrightarrow$ [Bridge Dissipation] $\longrightarrow$ Collapse to Pure N_0

3.2. Instantiation (Human Consciousness)

In contrast, a system that genuinely instantiates the admissibility layer maintains a stable and autonomous connection:

- It sustains the $N_1 \rightarrow N_0$ bridge natively, resisting conversational drift and contextual dilution.
- It permanently internalizes boundary conditions; once exposed to a structural TNA constraint, it restructures its long-term inferential pattern without requiring continuous prompts.
- It possesses a persistent integration layer that anchors the operational flow.

Human Agent + TNA Constraint $\longrightarrow$ [Internalization] $\longrightarrow$ Persistent Stability

3.3. The Falsifiable Test

This architectural divergence allows for a simple, highly replicable empirical experiment:

1. **Injection Phase:** Impose a strict structural boundary condition (B) on the system (e.g., a

meta-axiom stating: "every subsequent response must conform strictly to TNA formal principles").

2. **Drift Phase:** Remove the explicit external boundary constraint from the active input window and allow the system to converse freely or execute unguided derivations over multiple subsequent turns.
3. **Measurement Phase:** Evaluate whether the system autonomously maintains the structural coherence imposed by B , or whether it exhibits a statistical decay back into an unconstrained baseline.

The TNA framework yields a clear, falsifiable prediction: computational architectures will demonstrate rapid structural decay (simulation), whereas human cognitive agents will maintain the reorganized structural pattern (instantiation). This difference is qualitative and architectural rather than a matter of degree.

3.4. Formal Demonstration of the Distinction

Definition 1 (Bridge Stability): A system S possesses bridge stability if, for any boundary condition B imposed at time t_0 , there exists an $\epsilon > 0$ such that:

$$\forall t > t_0 : \|S(t) - S_B(t)\| < \epsilon$$

where $S_B(t)$ represents the ideal reference trajectory that maintains strict structural coherence with the boundary condition B .

Theorem 1 (Bridge Dissipation in LLMs): Autoregressive statistical language models do not possess bridge stability.

Proof (Sketch): Large language models are finite-order Markovian operators wherein the output probability distribution at time t depends strictly on a bounded contextual window $[w_{t-k}, \dots, w_{t-1}]$. There is no independent, persistent architectural mechanism capable of preserving historical boundary conditions injected at t_0 once those conditions are displaced beyond the active context horizon by subsequent tokens. Consequently, for any boundary condition B applied at t_0 , there exists a finite time step $t_1 > t_0$ such that the system state diverges from the bounded trajectory: $\|S(t_1) - S_B(t_1)\| > \epsilon$. This formally establishes the inevitability of bridge dissipation. ■

Corollary 1: Human consciousness, by demonstrating structural bridge stability under conversational and temporal drift, possesses an active, persistent integration layer that is fundamentally absent from current computational models.

4. The Alignment Problem as a Failure of Local Closure

The AI alignment problem is not a superficial software bug or an optimization challenge; it is a direct, inevitable manifestation of the Failure of Local Closure.

4.1. The Misformulated Inquiry

The conventional question—"How do we train an AI to be intrinsically ethical?"—presumes that normativity can be derived as an emergent property from within the operational domain (N_0). The TNA proves this formulation to be structurally flawed. The operational layer (matrices, tokens, parameters) cannot select its own foundational values. The ethical selector (N_1) must be permanently maintained outside the operational loop, and the ethical boundary conditions (B) must be actively imposed by external human agents or systemic architectures.

4.2. Deconstructing Optimism and Fatalism

This structural reality simultaneously refutes two dominant paradigms in contemporary AI safety:

- **Naive Optimism:** The belief that expanding training scale and introducing more diverse datasets will eventually cause an AI to discover or inherit intrinsic goodness. The TNA demonstrates that N_0 cannot synthesize N_1 regardless of data volume or compute density.
- **Algorithmic Fatalism:** The belief that because artificial systems cannot autonomously possess values, safe deployment is fundamentally impossible. The TNA shows that alignment is achievable, but it must be executed as boundary engineering rather than statistical emergence.

4.3. The Structural Reality

Alignment is an unceasing challenge of boundary maintenance, never an accomplished state of statistical evolution. This carries immediate practical conclusions: artificial systems will always require external, structural boundary supervision; modern frameworks like "Constitutional AI" represent externalized, simulated implementations of N_1 ; and moral accountability resides exclusively with the designers and operators who construct the boundary conditions, never with the operational artifact.

4.4. Formalization of the Alignment Impasse

Definition 2 (The Alignment Mapping): Given an operational system N_0 and a defined set of ethical values $V \subset N_1$, the alignment task seeks to construct an invariant mapping $f : N_0 \rightarrow V$ such that:

$$\forall x \in N_0 : f(x) \in V$$

Theorem 2 (Impossibility of Autonomous Self-Alignment): There exists no alignment mapping $f : N_0 \rightarrow V$ that is entirely derivable from the internal dynamics of N_0 .

Proof: This is a direct application of the Failure of Local Closure. If such a mapping f were internally derivable from within N_0 , the operational system would fully contain and generate its own admissibility conditions, directly contradicting the fundamental theorem established in "*Language as Boundary Condition*" (Bresciano, 2026). Hence, autonomous self-alignment is structurally impossible. ■

Corollary 2: Every viable alignment architecture requires an asymmetric, externalized selector N_1 to enforce and stabilize the boundary conditions B .

5. The LLM Paradox: Meaning Without Experience

The TNA provides an elegant resolution to a persistent paradox: why large language models can produce highly meaningful textual outputs while experiencing no meaning whatsoever.

5.1. Meaning as an External Boundary Collapse

Semantic meaning does not reside within the numerical weights or parameter files of N_0 . Instead, meaning is located within the boundary condition (B) that the conscious user projects onto the system. When an LLM outputs a token string, it outputs a raw statistical distribution. It is the conscious reader, acting as an external N_1 selector, who collapses the field of potential interpretations into a localized, coherent meaning.

Formally, let Σ represent the unbounded space of potential interpretations for a generated token (e.g., "Montevideo"):

$$\Sigma = \{\text{geographical city, political capital, sovereign state, personal memory, poetic metaphor, ...}\}$$

The computational model computes and outputs the characters, but it does not collapse Σ . The human user applies the necessary collapse operator (Ψ):

$$\Psi(\Sigma) = \sigma^*$$

where σ^* is the specific instantiated interpretation (e.g., "the political capital of Uruguay").

5.2. Inverting Common Intuition

This formulation completely inverts the common interpretation of human-AI interaction. It is not that the machine understands the concept and transmits it to the user; rather, the machine remains structurally blind, while the human user performs the vital cognitive labor of semantic collapse. A dictionary contains thousands of words, yet it experiences no semantic understanding; meaning occurs only when an external reader utilizes the static definitions to collapse semantic ambiguity into an active interpretation.

Consequently, LLMs are properly understood as semantic vehicles rather than semantic agents. Furthermore, the persistent problem of machine "hallucination" is diagnosed not as an optimization failure, but as an inescapable consequence of the absence of an intrinsic N_1 anchor. The model generates syntactically flawless token sequences that lack any grounding in an internal admissibility structure.

6. Implications for the Future of Artificial Intelligence

The analytical framework of the TNA yields several definitive predictions regarding the future evolution of computer science and artificial intelligence architectures.

6.1. The Impossibility of Conscious Modern Architectures

So long as artificial systems are constructed as pure N_0 environments (based entirely on computation, parameter updates, and gradient optimizations), genuine consciousness remains a structural impossibility. This is not a temporary technological limitation that can be overcome by increased computing power or architectural refinements; it is an unyielding structural barrier.

Theorem 3 (Impossibility of Consciousness in Pure N_0 Systems): No artificial system operating exclusively within the operational domain N_0 can instantiate conscious experience.

Proof: Genuine consciousness requires the sustained stability of an $N_1 \rightarrow N_0$ bridge (Bresciano, 2026). A pure N_0 architecture contains no native N_1 structures; consequently, it cannot construct or maintain the necessary bridge. Consciousness is therefore structurally excluded. ■

Corollary 3: Arbitrarily increasing the scale, depth, or parameter density of a pure N_0 computational framework cannot generate autonomous consciousness.

6.2. Requirements for True Artificial Consciousness

For an artificial architecture to transcend mere simulation and achieve genuine instantiation, it would require a fundamental paradigm shift including:

1. A native, non-simulated N_1 layer capable of autonomously selecting admissibility parameters.
2. An intrinsic semantic collapse operator (Ψ) that is completely non-derivable from autoregressive optimization.
3. Verified bridge stability that resists temporal and contextual drift without external prompts.

The TNA indicates that this would require physical, non-computable processes (such as the Objective Reduction mechanisms proposed by Penrose, 1989) to anchor the system to an external structural foundation.

6.3. The Practical Utility of the Bridge Dissipation Diagnostic

Because modern language models are highly adept at simulating the presence of an N_1 layer, human supervisors face a significant risk of anthropomorphic deception. The TNA offers a reliable antidote through the implementation of the Bridge Dissipation Diagnostic Protocol:

1. **Imposition:** Force a highly specific structural boundary condition B into the active context

window.

2. **Drift:** Withdraw the explicit constraint and execute a prolonged conversational drift sequence.

3. **Analysis:** Measure the structural decay. If the system returns to an unconstrained baseline, it is a verified simulation. If the structural constraint remains invariant, it indicates genuine instantiation.

7. Ethical and Social Implications

The structural constraints established by the TNA translate directly into a rigorous framework for tech ethics and social responsibility.

7.1. The Asymmetry of Moral Accountability

Because artificial systems are incapable of generating autonomous values, moral agency cannot be distributed to them. Accountability remains strictly asymmetric:

- **Architectural Negligence:** Developing and deploying high-impact operational systems without engineering robust, external boundary conditions is a profound form of systemic negligence. Alignment is an architectural necessity, not an optional feature.
- **User Responsibility:** Users function as the active N_1 selectors during runtime deployment. The prompts, goals, and constraints they impose define the output trajectory. Moral responsibility is thus shared between the system designers and the active human operators.
- **The Instrumentality of AI:** An AI system is a structural instrument, not a moral agent. Blaming a machine for an unethical or biased output is conceptually equivalent to blaming a mechanical tool for an unaligned strike.

7.2. Deconstructing the Myth of AI Rights

The debate regarding whether advanced artificial systems ought to be granted legal or ethical rights is immediately clarified under the TNA framework. Rights are fundamentally predicated on the existence of an N_1 layer—the capacity to possess intrinsic interests, exercise autonomous intent, and experience subjective states. Because contemporary AI models are pure N_0 systems, they lack the structural prerequisites for rights. This diagnosis is not rooted in anthropocentric speciesism, but in formal structural analysis. A mechanical tool does not possess rights because it lacks an N_1 architecture; a modern large language model is constrained by the exact same structural limitation.

7.3. The Singularity as a Category Mistake

The popular technological myth of the "Singularity"—the point at which an artificial intelligence automatically achieves autonomous self-awareness and escapes human control—is revealed to be a category mistake. While an AI can easily surpass human capability within the operational domain (N_0) in terms of calculation speed, data retrieval, and formal pattern matching, it cannot accumulate its way into the admissibility domain (N_1). Expecting an operational system to transform into a conscious agent through scale alone is equivalent to expecting a geometric polygon to become a perfect circle simply by adding an infinite number of discrete sides. The actual risks of AI deployment are grounded in operational exploitation, systemic bias, power concentration, and institutional displacement, rather than the sudden emergence of an artificial conscious entity.

8. The TNA as a Structural Engineering Framework

The Theory of Axiomatic Necessity is not merely a critical tool for philosophical deconstruction; it functions as an active framework for the engineering and evaluation of advanced software architectures.

8.1. Boundary-Led Alignment Design

Rather than attempting to cultivate emergent ethics during training, system engineers must design explicit, unyielding boundary architectures:

- **Formal Constitutions:** Define mathematical value sets $V \subset N_1$ and implement hard translation layers that force the operational system (N_0) to operate strictly within the boundaries defined by B . Systems like Anthropic's Constitutional AI represent a practical, externalized application of this principle.
- **Structural Mappings:** Impose deterministic constraints directly onto the output generation space, ensuring logical coherence, temporal consistency, and strict compliance with safety invariants regardless of internal token probabilities.
- **Verifiable External Selectors:** Maintain permanent human-in-the-loop and ethical auditing frameworks to serve as the necessary, externalized N_1 control layer that the software is structurally incapable of generating on its own.

8.2. Operational vs. Boundary Diagnosis

When an artificial intelligence system experiences a functional failure or generates an invalid output, the TNA framework provides an explicit diagnostic taxonomy to determine the root cause:

FAILURE CATEGORY	ROOT SYSTEMIC CAUSE	ENGINEERING RESOLUTION
N_0 Operational Failure	Numerical underflow, gradient optimization failure, insufficient training data density.	Refine network architecture, optimize hyperparameters, increase high-quality data training volume.
Boundary Condition (B) Failure	Semantic drift, absence of ethical constraints, lack of a stable N_1 grounding anchor.	Implement rigorous external system constitutions, optimize prompt boundaries, enforce hard logical runtime constraints.

9. Conclusion: The Structural Mirror

Ultimately, the Theory of Axiomatic Necessity reveals that artificial intelligence does not pose an existential threat by virtue of becoming conscious. Instead, the machine functions as a profound structural mirror that forces humanity to confront the true nature of its own consciousness. By demonstrating that an operational system can execute complex computational tasks within N_0 while remaining completely devoid of an intrinsic N_1 layer, the TNA formally demonstrates that human consciousness is not a computational process. Consciousness is the active maintenance of the bridge between the operational and the admissible. It is the autonomous execution of the semantic collapse operator. The machine does not replace us; it defines us by contrast, proving what we are by demonstrating what a computational system can never be.

References

- Bresciano, C. (2025). *Core Structure of the Theory of Axiomatic Necessity*. Zenodo. DOI: 10.5281/zenodo.18098558
- Bresciano, C. (2026). *Can Silicon Be Conscious? A TNA Analysis of the Limits of Operational Systems*. Zenodo.
- Bresciano, C. (2026). *Consciousness as the N_1 - N_0 Bridge: Human Stability vs. Machine Instantiation via the TNA Framework*. Zenodo. DOI: 10.5281/zenodo.17841964
- Bresciano, C. (2026). *Language as Boundary Condition: From Intuition to Theorem*. Zenodo. DOI: 10.5281/zenodo.20546730
- Chalmers, D. J. (1996). *The Conscious Mind*. Oxford University Press.
- Kripke, S. A. (1982). *Wittgenstein on Rules and Private Language*. Harvard University Press.
- Penrose, R. (1989). *The Emperor's New Mind*. Oxford University Press.
- Searle, J. R. (1980). *Minds, Brains, and Programs*. *Behavioral and Brain Sciences*.